

PrimeSim™ HSPICE® Quick Start Guide

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About the PrimeSim HSPICE Quick Start Guide

The *PrimeSim HSPICE Quick Start Guide* provides information to quickly get you started with using the Synopsys PrimeSim HSPICE tool for simulation and analysis of your circuit designs.

It provides an overview of the PrimeSim HSPICE simulation flow and information on the installation, invocation, and documentation of the PrimeSim HSPICE tool. It also provides links to the detailed documentation for the analyses, elements, and models supported by the PrimeSim HSPICE tool.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeSim HSPICE Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeSim HSPICE simulator, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Cadence® Virtuoso® Analog Design Environment

Synopsys PrimeWave™ Design Environment

Synopsys SolvNetPlus support site

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

PrimeSim HSPICE Overview

Provides an overview of the PrimeSim HSPICE simulation flow, analysis types, element statements, and supported semiconductor device models. Describes the PrimeSim HSPICE documentation set and how to access the documentation.

This chapter discusses the following topics:

- [Simulation Flow](#)
- [Analysis Types](#)
- [Elements](#)
- [Models](#)
- [Demonstration Files](#)
- [Documentation](#)
- [PrimeSim HSPICE Application Notes](#)

Simulation Flow

[Figure 1 on page 9](#) illustrates the PrimeSim HSPICE simulation flow.

In a PrimeSim HSPICE simulation, the following sequence of events occurs

- Invocation:

You invoke the PrimeSim HSPICE tool by entering the following command, for example, at the shell prompt:

```
% hspice -i demo.sp -o demo.lis &
```

This command invokes the PrimeSim HSPICE tool with the input netlist file `demo.sp` and writes the output listing to the file `demo.lis`. The `&` character at the end of the command invokes the PrimeSim HSPICE tool in the background, so that you can continue to use the command window and the keyboard while the PrimeSim HSPICE tool runs.

Figure 1

- Script execution:

The `hspice` command starts the PrimeSim HSPICE executable from the appropriate architecture (machine type) directory. The UNIX `run` script launches a PrimeSim HSPICE simulation. This procedure establishes the environment for the PrimeSim HSPICE executable. The script prompts for information such as the platform that you are using and the PrimeSim HSPICE version to run. (The available versions of the PrimeSim HSPICE tool are pre-determined during the installation of the tool.)

- Licensing:

The PrimeSim HSPICE tool supports the FLEXlm licensing management system. When you use FLEXlm licensing, the PrimeSim HSPICE tool reads the `SNPSLMD_LICENSE_FILE` or the `LM_LICENSE_FILE` environment variable to find the available license server. If the PrimeSim HSPICE tool cannot authorize the access, the job terminates at this point and the PrimeSim HSPICE tool prints an error message in the output listing file.

- Simulation configuration:

The PrimeSim HSPICE tool reads the appropriate `hspice.ini` file. The search order for the configuration file is:

1. The current working directory.
2. The \$HOME directory of the user.
3. The PrimeSim HSPICE installation directory.

- Design input:

The PrimeSim HSPICE tool opens the input netlist file `demo.sp`. If this file does not exist, a `No input data` error message appears in the output listing file.

The PrimeSim HSPICE tool then opens the output listing file `demo.lis` for writing. If you do not own the current directory, the tool terminates with an error message.

- Library input:

The PrimeSim HSPICE tool reads the files, if any, that you specified in the `.INCLUDE` and `.LIB` statements.

- Operating point initialization:

The PrimeSim HSPICE tool reads initial conditions, if any, that you specified in the `.IC` and `.NODESET` commands, finds an operating point (that you can save with a `.SAVE` command), and writes any operating point information that you requested.

- **Analysis:**

The PrimeSim HSPICE tool performs the specified single or multipoint sweep of the design and produces a set of output files.

- **.ALTER commands:**

Using `.ALTER` commands, you can vary the simulation conditions and repeat the specified single or multipoint analysis. You can activate multiprocessing while running `.ALTER` cases by entering `hspice -dp` on the command line.

- **Suspending a simulation:**

To suspend a simulation job, press Ctrl+Z. The Load Sharing Facility (LSF) then frees up the PrimeSim HSPICE license for another simulation job. To resume the suspended job, on the same terminal, type either `fg` or `bg`. This command accesses an PrimeSim HSPICE license and continues the suspended simulation.

- **Normal termination:**

After you complete the simulation, the PrimeSim HSPICE tool closes the opened files and releases the license tokens used for the simulation.

Analysis Types

The PrimeSim HSPICE tool can simulate electrical circuits in the time and frequency domains. The PrimeSim HSPICE tool provides analysis capabilities that support the design of advanced analog and high-speed circuits. See the [Table 1](#) for the analysis types supported by the PrimeSim HSPICE tool. See also [Chapter 3, Running an Analysis](#), in this guide.

Table 1 PrimeSim HSPICE Analysis Types

Analysis	Description
<code>.AC</code>	AC sweep analysis
<code>.ACMATCH</code>	Analyzes the effects of local variations on the AC response of a circuit.
<code>.ACPHASENOISE</code>	AC phase noise analysis
<code>.BIASCHK</code>	Dynamic bias checking
<code>.DC</code>	DC sweep analysis
<code>.DCMATCH</code>	Analyze the effects of local variations on the DC characteristics of a circuit.

Table 1 *PrimeSim HSPICE Analysis Types (Continued)*

Analysis	Description
.DCSENS	DC sensitivity analysis using variation definitions specified using variation blocks.
.DISTO	Distortion analysis
.ENV	Envelope analysis
.ENVFFT	Envelope Fast Fourier Transform (FFT) analysis
.ENVOSC	Envelope analysis for oscillators
.FFT	FFT analysis
.FOUR	Fourier analysis
.HB	Harmonic balance analysis
.HBAC	Multi-tone harmonic balance AC analysis
.HBLIN	Frequency translation S-parameter extraction
.HBLSP	Large-signal S-parameter analysis
.HBNOISE	Multi-tone harmonic balance noise analysis
.HBOSC	Harmonic balance analysis for oscillators
.HBXF	Harmonic balance transfer function analysis
.LIN	Linear network analysis
.LSTB	Loop stability analysis
.MOSRA	MOSFET reliability analysis
.NOISE	Small-signal noise analysis
.OP	Operating point analysis
.PHASENOISE	Phase noise analysis
.PTDNOISE	Periodic time-domain noise analysis
.PZ	Pole/zero analysis
.SAMPLE	Data sampling noise analysis

Table 1 *PrimeSim HSPICE Analysis Types (Continued)*

Analysis	Description
.SENS	Sensitivity analysis. Determines DC small-signal sensitivities of output variables for circuit parameters.
.SN	Shooting Newton analysis
.SNAC	Shooting Newton AC analysis
.SNFT	Shooting Newton discrete Fourier transform analysis
.SNNOISE	Shooting Newton noise analysis
.SNOSC	Shooting Newton analysis for oscillators
.SNXF	Shooting Newton transfer function analysis
.STATEYE	Statistical eye diagram analysis
.TF	Transfer function analysis. Calculates small-signal values for transfer functions for both DC and AC simulations.
.TRAN	Transient analysis
.TRANNOISE	Transient noise analysis

See also [Chapter 3, Running an Analysis](#), in this guide.

Elements

PrimeSim HSPICE element statements describe the netlists of devices and sources. Each element begins with a unique letter that identifies its device type. See the [Table 2](#). See also [Chapter 4, Working With Elements](#) in this guide.

Table 2 PrimeSim HSPICE Elements

Element	Description
B	IBIS Buffer
C	Capacitor
D	Diode
E	Voltage Dependent Voltage Source
F	Current Dependent Current Source
G	Voltage Dependent Current Source
H	Current Dependent Voltage Source
I	Independent Current Source
J	JFET or MESFET
K	Mutual Inductor
L	Inductor
M	MOSFET
P	Port
Q	BJT Transistor
R	Resistor
S	S Parameter
T	Ideal Transmission Line
U	Lossy Transmission Line
V	Independent Voltage Source
W	Lossy Transmission Line (preferred)
X	Subcircuit

See also [Chapter 4, Working With Elements](#) in this guide.

Models

PrimeSim HSPICE tool supports many common semiconductor device models. See the following table. See also [Chapter 5, Working With Models](#) in this guide.

Table 3 Device Models Supported by the PrimeSim HSPICE Tool

Device Model	Description
Capacitors	
CMC Varactor	Level=7 CMC Varactor Model
Resistors	
CMC R2 Resistor	Level=2 CMC R2 resistor model
CMC R3 Resistor	Level=5 CMC R3 resistor model
Diodes	
Non-Geometric Junction Diode	Level=1 non-geometric junction diode model
Fowler-Nordheim Diode	Level=2 Fowler-Nordhiem diode model
Geometric Junction Diode	Level=3 geometric junction diode model
JUNCAP1 Diode	Level=4 junction capacitance diode model
Philips D500 Diode	Level=5 Philips D500 advanced diode mode
JUNCAP2 Diode	Level=6 junction capacitance diode modell
CMC Diode	Level=7 CMC diode model
JFETs and MESFETs	
Level 1 JFET	Level=1 JFET model
Level 2 JFET	Level=2 JFET model
Level 3 MESFET	Level=3 MESFET model
TriQuint MESFET	Level=7 TriQuint MESFET model

Table 3 *Device Models Supported by the PrimeSim HSPICE Tool (Continued)*

Device Model	Description
Materka MESFET	Level=8 Materka MESFET
BJTs	
Level 1 BJT	Level=1 BJT model
Level 2 BJT	Level=2 BJT model
VBIC	Level=4 VBIC Model
Philips Bipolar MEXTRAM Level 503	Level=6 Philips Bipolar Model
LEVEL=6 Philips Bipolar Model (MEXTRAM)	Level=6 Philips Bipolar Model
HICUM	Level=8 HICUM Model
VBIC99	Level=9 VBIC99 Model
Philips MODELLA Bipolar	Level=10 Philips MODELLA Bipolar Model
UCSD HBT	Level=11 UCSD HBT Model
HICUM/L0	Level=13 HICUM/L0 Model
MOSFETs	
Schichman-Hodges	Level = 1 Schichman-Hodges Model
MOS2 Grove-Frohman (SPICE 2G)	Level=2 MOS2 Grove-Frohman Model
MOS3 Empirical (SPICE 2G)	Level=3 MOS3 Empirical Model
Grove-Frohman Level 2	Level=4 Grove-Frohman Model
AMI-ASPEC	Level=5 AMI-ASPEC Model
Lattin-Jenkins-Grove (ASPEC)	Level=6 Lattin-Jenkins-Grove (ASPEC) Model
Lattin-Jenkins-Grove (SPICE)	Level=7 Lattin-Jenkins-Grove (SPICE) Model
Advanced Level 2 MOS2 Grove-Frohman (SPICE 2G)	Level=8 Advanced Level 2: MOS2 Grove-Frohman Model (SPICE 2G)
BSIM	Level=13 BSIM Model

Table 3 *Device Models Supported by the PrimeSim HSPICE Tool (Continued)*

Device Model	Description
SOSFET	Level=27 SOSFET Model
BSIM Derived	Level=28 BSIM Derived Model
Cypress Depletion	Level=38 Cypress Depletion Model
BSIM2	Level=39 BSIM2 Model
HP a-Si TFT	Level=40 HP a-Si TFT Model
BSIM3 Version 2 MOSFET	Level=47 BSIM3 Version 2 MOSFET Model
BSIM3 Version 3 MOSFET	Level=49 and Level=53 BSIM3 Version 3 MOSFET Model
Philips MOS9	Level=50 Philips MOS9 Model
BSIM4 MOSFET	Level=54 BSIM4 MOSFET Model
EPFL-EKV MOSFET	Level=55 EPFL-EKV MOSFET Model
BSIM3 SOI	Level=57 BSIM3 SOI Model
UFSOI MOSFET	Level=58 UFSOI MOSFET Model
BSIM3 SOI FD	Level=59 BSIM3 SOI FD Model
BSIM3 SOI DD	Level=60 BSIM3 SOI DD Model
RPI a-Si TFT	Level=61 RPI a-Si TFT Model
RPI Poli-Si TFT	Level=62 RPI Poli-Si TFT Model
Philips MOS11	Level=63 Philips MOS11 Model
STARC HiSIM MOSFET	Level=64 STARC HiSIM MOSFET Model
SSIMSOI	Level=65 SSIMSOI Model
PrimeSim HSPICE HVMOS	Level=66 PrimeSim HSPICE HVMOS Model
STARC HiSIM2 MOSFET	Level=68 STARC HiSIM2 MOSFET Model
PSP	Level=69 PSP Model

Table 3 *Device Models Supported by the PrimeSim HSPICE Tool (Continued)*

Device Model	Description
BSIMSOI4	Level=70 BSIMSOI4 Model
TFT	Level=71 TFT Model
BSIM-CMG MOSFET	Level=72 BSIM-CMG MOSFET Model
HiSIM_HV	Level=73 HiSIM_HV Model
MOS Model 20	Level=74 MOS Model 20 Model
LETI-UTSOI MOSFET	Level=76 LETI-UTSOI MOSFET Model
BSIM6 MOSFET	Level=77 BSIM6 MOSFET Model
BSIM-IMG	Level=78 BSIM-IMG Model
EKV3	Level=79 EKV3 Model

See also [Chapter 5, Working With Models](#) in this guide.

Demonstration Files

The PrimeSim HSPICE tool is shipped with as many as 350 demonstration netlist files that illustrate the use of PrimeSim HSPICE commands and control options, for various applications.

You can find the demonstration files at:

- For PrimeSim HSPICE: `$installdir/demo/hspice/`
- For PrimeSim HSPICE advanced analyses: `$installdir/demo/hspice/rf_examples/`

Where `$installdir` is the directory where the PrimeSim HSPICE tool is installed on your network or machine.

For detailed listing of the PrimeSim HSPICE demonstration file categories and descriptions of the netlists, see:

- For PrimeSim HSPICE examples: Listing of Demonstration Input Files in the *PrimeSim HSPICE User Guide: Demonstration Netlists*.
- For PrimeSim HSPICE advanced analyses: Advanced Analog Demonstration Input Files in the *PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis*.

Documentation

The documentation set for the PrimeSim HSPICE tool is available as PDF manuals and online Help.

The following sections discuss these topics:

- [The PrimeSim HSPICE Documentation Set](#)
- [Finding PrimeSim HSPICE Documentation](#)
- [Accessing PrimeSim HSPICE Documentation](#)
- [Launching Context-Sensitive Help](#)

The PrimeSim HSPICE Documentation Set

The PrimeSim HSPICE documentation set, both PDF and online Help, includes the following manuals:

Manual	Description
PrimeSim HSPICE® User Guide: Demonstration Netlists	Provides detailed information on locating and running the PrimeSim HSPICE demonstration netlists.
PrimeSim HSPICE Quick Start Guide	Provides information to quickly get you started with using the PrimeSim HSPICE tool for simulation and analysis of your circuit designs.
<i>PrimeSim HSPICE® User Guide: Basic Simulation and Analysis</i>	Describes how to use PrimeSim HSPICE to simulate and analyze your circuit designs, and includes simulation applications. This is the main PrimeSim HSPICE user guide.
<i>PrimeSim™ Continuum User Guide: Elements</i>	Describes the syntax for the basic elements of a circuit netlist in PrimeSim HSPICE, descriptions of each of the elements keywords, and examples of common usage for each element.
<i>PrimeSim HSPICE® User Guide: Signal Integrity Modeling and Analysis</i>	Describes how to use PrimeSim HSPICE to achieve and maintain signal integrity in your chip design.
<i>PrimeSim HSPICE® User Guide: Advanced Analog Simulation and Analysis</i>	Describes how to use a special set of analysis and design capabilities that PrimeSim HSPICE to support RF and high-speed circuit design.
<i>PrimeSim™ AVA User Guide</i>	Describes the principles of HSMC and Sigma Amplification simulations and provides information on how to perform an HSMC and Sigma Amplification simulation in PrimeSim HSPICE.

Manual	Description
<i>PrimeSim™ MOSRA User Guide</i>	Describes how to set up MOS reliability analysis (MOSRA) and the MOSRA simulation flow.
<i>PrimeSim™ Continuum Reference Manual: Commands and Control Options</i>	Describes the PrimeSim HSPICE commands you can use to simulate and analyze your circuit designs.
<i>PrimeSim™ Continuum Reference Manual: Device Models</i>	Describes standard models you can use when simulating your circuit designs in PrimeSim HSPICE, including passive devices, diodes, JFET and MESFET devices, and BJT devices.
<i>PrimeSim™ Continuum Reference Manual: MOSFET Models</i>	Describes available MOSFET models you can use when simulating your circuit designs in PrimeSim HSPICE.
<i>HSPICE Integration to Cadence® Virtuoso® Analog Design Environment User Guide</i>	Describes use of the PrimeSim HSPICE simulator integration to the Cadence tool.

Finding PrimeSim HSPICE Documentation

You can find the PrimeSim HSPICE documentation files, the online Help files and the PDFs, in your PrimeSim HSPICE installation. The documentation files are in the `$installdir/docs_help` directory, where `$installdir` is the directory where the PrimeSim HSPICE tool is installed on your network or machine.

You will find the PDFs in the `$installdir/docs_help/ni` directories.

Accessing PrimeSim HSPICE Documentation

You can view the latest version of the PrimeSim HSPICE online help or download the documentation in PDF format from the SolvNetPlus online support site at [PrimeSim HSPICE Documentation on SolvNet](#).

To view and print the PrimeSim HSPICE documentation in PDF, Synopsys recommends that you install Adobe Reader on your machine. For best results, use Adobe Reader version 6 or later.

You can access the PrimeSim HSPICE documentation in PDF format from the command line using the following command:

```
% hspice -doc
```

To launch the searchable, HTML browser-based PrimeSim HSPICE online help system, enter the following command on the command line:

```
% hspice -help
```

You must have an HTML browser installed on your machine to launch and view the online help.

Launching Context-Sensitive Help

The PrimeSim HSPICE online help is context sensitive. That is, you can directly access and view information about PrimeSim HSPICE commands, options, keywords, and other common help topics in the online help.

- For information about a command, use:

```
% hspice -help command
```

For example:

```
% hspice -help .AC
```

- For information about a control option, use:

```
% hspice -help .OPTION_option
```

For example:

```
% hspice -help .OPTION_DELMAX
```

- For information about a demo file, use:

```
% hspice -help demo_file_name
```

For example:

```
% hspice -help clockbuf4sn.sp
```

- For information about a topic, use:

```
% hspice -help keyword
```

For example:

```
% hspice -help bsim3v3
```

For the list of keywords for online help topics, see PrimeSim HSPICE Simulation Command Reference and PrimeSim HSPICE Simulation Control Options Reference in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

PrimeSim HSPICE Application Notes

To obtain the PrimeSim HSPICE application notes, go to the SolvNetPlus online support site at <https://solvnetplus.synopsys.com> or the PrimeSim HSPICE web site at <https://www.synopsys.com/verification/ams-verification/hspice.html>.

Note:

You must be logged in to SolvNetPlus for the link to connect directly to the article. If you are prompted to log in to SolvNetPlus upon clicking the link to the article, log in, then click the link again to reach the article.

2

Customizing Your Environment

Presents an overview of PrimeSim HSPICE installation, licensing, and environment settings.

To obtain the latest PrimeSim HSPICE installation notes, go to <http://www.synopsys.com/install>. For detailed licensing setup and troubleshooting assistance, see the *Synopsys Licensing QuickStart Guide* at <http://www.synopsys.com/licensing>.

This chapter discusses the following topics:

- [Installing the PrimeSim HSPICE Tool—An Overview](#)
- [Setting Up the Licensing Environment](#)

Installing the PrimeSim HSPICE Tool—An Overview

The PrimeSim HSPICE tool is available for the Red Hat Enterprise Linux (RHEL), SUSE Linux Enterprise Server, Windows, and ARM64 platforms. For detailed information on the supported compute platforms, operating systems, and windowing environments for your PrimeSim HSPICE release, see *Synopsys® PrimeSim HSPICE® Installation Notes*.

Installing the PrimeSim HSPICE Tool on UNIX

The PrimeSim HSPICE tool uses the Synopsys Installer, which allows you to install the product using either a text script or a graphical user interface (GUI). For information about downloading the Synopsys Installer, see the Synopsys Installer section at <http://www.synopsys.com/install>.

To install the PrimeSim HSPICE tool, follow the procedures described in the *Synopsys Installation Guide* downloaded from this site.

For detailed information, see *Synopsys® PrimeSim HSPICE® Installation Notes*.

Installing the PrimeSim HSPICE Tool on Windows

For installation on Windows, download the appropriate PrimeSim HSPICE release, from the [SolvNetPlus Download Center](#), to a temporary directory. Double-click the downloaded HSPICE.exe file to start the installation process. For detailed information, see *Synopsys® PrimeSim HSPICE® Installation Notes*.

Setting Up the Licensing Environment

The PrimeSim HSPICE tool requires you to set the `LM_LICENSE_FILE` environment variable. This variable specifies the location of the `license.dat` license file. Set the `LM_LICENSE_FILE` environment variable to `port@hostname` to point to a license file on a server.

- If you are using the C shell, add the following line to the `.cshrc` file:

```
setenv LM_LICENSE_FILE port@hostname
```

- If you are using the Bash or Bourne shell, add these lines to the `.bashrc` or the `.profile` file:

```
LM_LICENSE_FILE=port@hostname  
export LM_LICENSE_FILE
```

The `port` and `hostname` variables correspond to the TCP port and the license server host name specified in the `SERVER` line of the Synopsys license file.

Each license file can contain licenses for many packages from multiple vendors. You can specify multiple license files by separating each entry. For Linux platforms, use a colon (:) as the separator. For Windows platforms, use a semicolon (;) as the separator.

For detailed information on setting up the user environment for PrimeSim HSPICE licensing, see *Synopsys® PrimeSim HSPICE® Installation Notes*.

Configuring the PrimeSim HSPICE License

You must set up one of the following environment variables to point to the license paths. When specifying the pointer to the license paths, use the full path of the license file, the `port@host` syntax, or a combination of the two syntax types separated by colon.

- `SNPSLMD_LICENSE_FILE`

A pointer to the Synopsys license paths

- `LM_LICENSE_FILE`

A pointer to all license paths. If you specify both environment variables, `SNPSLMD_LICENSE_FILE` takes precedence over `LM_LICENSE_FILE`.

In C-Shell:

```
setenv SNPSLMD_LICENSE_FILE pointer_to_license_paths
```

In Bourne, Korn, or Bash shells:

```
export SNPSLMD_LICENSE_FILE=pointer_to_license_paths
```

PrimeSim HSPICE supports two license modes as described in the following sections:

- [Controlling the License Queuing Interval](#)
- [Configuring the PrimeSim Continuum Token-Based License Mode](#)

Controlling the License Queuing Interval

As described in the previous section, setting the environment variable `PRIMEISM_WAIT_LICENSE` causes PrimeSim HSPICE to wait for an available license from the license server. If no license becomes available, then PrimeSim HSPICE waits for 2 minutes by default. There may be times where it is desirable to set different queueing time, such as during scripted operations.

To do this, you can set the environment variable: `PRIMESIM_WAIT_LICENSE_TIMEOUT`. The argument to `PRIMESIM_WAIT_LICENSE_TIMEOUT` is in minutes.

For example:

```
#!/bin/csh -f
setenv PRIMESIM_WAIT_LICENSE 1
setenv PRIMESIM_WAIT_LICENSE_TIMEOUT 60
hspice -i myjob.sp -o outfile
```

Here, `PRIMESIM_WAIT_LICENSE` enables PrimeSim HSPICE to wait for an available license. Next, the `PRIMESIM_WAIT_LICENSE_TIMEOUT` is set to 60 minutes. The license request de-queues if no license is available in 60 minutes and the script can continue. This feature is available on all platforms.

Configuring the PrimeSim Continuum Token-Based License Mode

To enable the PrimeSim Continuum token-based license mode, use one of the following methods:

- Define the shell environment variable `PRIMESIM` to 1.

In C-Shell:

```
setenv PRIMESIM 1
```

In Bash shell:

```
export PRIMESIM=1
```

- Use the `-primesim 1` command line option.

If the shell environment variable is set and the `-primesim` command line option is used, the command line option will take precedence and override the shell environment variable setting.

For example,

```
% setenv PRIMESIM 1
% hspice netlist.sp
```

Or

```
% hspice netlist.sp -primesim 1
```

The following table lists the shell environment variables that control the license mode and license queuing.

Table 4 *Variables for Controlling Token-Based License Mode and License Queuing*

Environment Variable	Description
PRIMESIM 0 1 2	This variable enables PrimeSim Continuum token-based license mode. Valid values: • 0: Disables token-based license mode. This is the default. • 1: Enables token-based license mode. • 2: (Default) Enables standalone and token-based license mode. By default, the standalone keys are attempted first (see <code>PRIMESIM_ORDER</code>). This variable can also be enabled via the <code>-primesim</code> command line option. Note: Arguments are mandatory for <code>-primesim</code> and <code>-primesim_order</code>. If no value specified, the errors are like: • <code>*error* lack of value [0 1 2] specified for -primesim.</code> • <code>*error* lack of value [0 1 2] specified for -primesim_order.</code>

Table 4 *Variables for Controlling Token-Based License Mode and License Queuing*
(Continued)

Environment Variable	Description
<code>PRIMESIM_ORDER 0 1</code>	This variable controls whether standalone keys or token-based keys are attempted first when the <code>PRIMESIM</code> variable is set to 2. • 0: (Default) Standalone keys are attempted first. • 1: Token-based keys are attempted first. Note: Arguments are mandatory for <code>-primesim</code> and <code>-primesim_order</code>. If no value specified, the errors are like: • *error* lack of value [0 1 2] specified for <code>-primesim</code>. • *error* lack of value [0 1 2] specified for <code>-primesim_order</code>.
<code>PRIMESIM_WAIT_LICENSE 1 0</code>	This variable controls license queuing. • 0: Disables license queuing. • 1: (Default) Enables license queuing.
<code>PRIMESIM_WAIT_LICENSE_TIMEOUT value</code>	This variable controls the timeout period for license queuing. The value is in minutes and cannot exceed 43,200 minutes (30 days). The default period is 2 minutes. Non-integer values or negative integer values will result in an error. The following example sets the queue timeout period to 30 minutes: <pre>% setenv PRIMESIM_WAIT_LICENSE_TIMEOUT 30</pre>
<code>PRIMESIM_WAIT_LICENSE_INTERVAL value</code>	This variable controls the timeout period of sleeping during license queuing. The value is in seconds and cannot exceed 60 seconds and cannot be less than 1 second. The default period is 30 seconds. Integer values beyond [0, 60] will result in a warning message with resetting into the default 30 seconds. The following example sets the queue timeout period to 2 seconds: <pre>% setenv PRIMESIM_WAIT_LICENSE_LICENSE 2</pre>

Table 4 *Variables for Controlling Token-Based License Mode and License Queuing*
(Continued)

Environment Variable	Description
<code>PRIMESIM_ONCE_CHECKOUT_LIC 1</code>	By default, when using token-based license mode, a queued simulation job checks out license tokens incrementally, as they become available, until the job has enough license tokens to run. Set this environment variable to 1 so that a queued simulation job waits until enough license tokens are available to run before claiming them all at once. Note: This environment variable is used only by the PrimeSim and PrimeSim HSPICE simulators and is ignored by the PrimeSim XA simulator.

Note:

The `META_QUEUE` and `META_QUEUE_TIME` environment variables are obsolete since S-2021.09 release. They are replaced with `PRIMESIM_WAIT_LICENSE` and `PRIMESIM_WAIT_LICENSE_TIMEOUT` appropriately.

Configuring the Stand-Alone License Mode

The standalone license mode is the default mode.

Setting Up License Queuing Variable

The optional `PRIMESIM_WAIT_LICENSE` environment variable causes PrimeSim HSPICE to wait for an available license. It is particularly helpful in environments where the tool runs sequentially from batch files and a license checkout failure could result in the loss of important data.

To enable queueing of PrimeSim HSPICE licenses, set:

```
setenv PRIMESIM_WAIT_LICENSE 1
```

If you have five PrimeSim HSPICE or advanced analog feature floating licenses and have checked out all five licenses, then with the `PRIMESIM_WAIT_LICENSE` environment variable enabled, the next job submitted waits in the queue until a license is available. When you have enabled `PRIMESIM_WAIT_LICENSE` and all available licenses are in use, PrimeSim HSPICE issues another message advising that no licenses are available.

When there are multiple license servers, the PrimeSim HSPICE tool queries all of the servers listed in the `LM_LICENSE_FILE` variable and, if no licenses are available, queues the jobs on the servers, when `PRIMESIM_WAIT_LICENSE` is enabled.

For example:

```
LM_LICENSE_FILE = 27000@server1;27000@server2;27000@server3
```

3

Running an Analysis

Contains information on running an analysis in the PrimeSim HSPICE tool. Introduces the PrimeSim HSPICE netlist structure and the analysis outputs. Presents a few simulation examples.

This chapter discusses the following topics:

- [Running the PrimeSim HSPICE Tool](#)
- [Netlist Structure](#)
- [Output Files](#)
- [Simulation Examples](#)

Running the PrimeSim HSPICE Tool

Use the following syntax to run the PrimeSim HSPICE tool:

```
hspice [-dpredict]
        [-i | -hspice path/input_file]
        [-o path/output_file]
        [-t $TSTOP]
        [-n number]
        [-gz]
        [-d]
        [-C path/input_file]
        [-CC path/input_file]
        [-I]
        [-K]
        [-L command_file]
        [-include_first ins_filename]
        [-include_last app_filename]
        [-S]
        [-case 0|1]
        [-datamining -i datamining.cfg [-o outname]]
        [-dp [num]
            [-dpconfig dp_configuration_file]
            [-dplocation NFS|TMP]
            [-merge]
```

```
[-dpgui]
[-dpmode alter|sweep]
[-dpincremental original_path]
[-dscale output.dynamic] ]
[-mp process_count]
[-mt thread_count]
[-hpp]
[-meas measure_file]
[-mrasm [0|1|2|3]]
[-top subcktname]
[-restore checkpoint_file/-auto_restore]
[-hdl file_name]
[-hdlpath pathname]
[-vamodel name]
[-vamodel name2...]
[-sae]
[-search | -inc path1[:path2:path3...]]
[-alter_select num | num1:num2 | alter_name]
[-help | -webhelp]
[-doc]
[-h]
[-v | -version]
[-t | time_value]
[-wavefmt | -format fsdb|wdf|psf|tr0]
```

For a detailed description of the `hspice` command syntax and arguments, see `hspice` in the Invoking PrimeSim HSPICE Simulations *PrimeSim™ HSPICE® User Guide: Basic Simulation and Analysis*. Examples for starting and running the PrimeSim HSPICE tool, see [Starting PrimeSim HSPICE — Examples](#).

For multiple processing, multithreading, distributed processing, and PrimeSim HSPICE Precision Parallel features, see Chapter 4, Distributed Processing, Multithreading, and PrimeSim HSPICE Precision Parallel in *PrimeSim HSPICE® User Guide: Basic Simulation and Analysis*.

For the analysis types supported by the PrimeSim HSPICE tool, see [Analysis Types on page 11](#) in this guide.

Netlist Structure

The following sections discuss these topics:

- [PrimeSim HSPICE Netlists](#)
- [Sections in a Netlist File](#)
- [Examples](#)

PrimeSim HSPICE Netlists

PrimeSim HSPICE netlists can be generated from text editors or from netlisters that generate circuits from schematics. The PrimeSim HSPICE tool can accept either hierarchical or flat netlists.

The process of creating a schematic involves the following steps

1. Symbol creation with a symbol editor.
2. Circuit encapsulation.
3. Property creation.
4. Symbol placement.
5. Symbol property definition.
6. Wire routing and definition.

Sections in a Netlist File

The following table lists and describes the sections in a PrimeSim HSPICE netlist:

Table 5 *Sections of the Input Netlist File*

Section	Command/Control Option Used	Description
Title	.TITLE	The first line in the netlist is the title of the input netlist file. The .TITLE command is optional.
Setup	.OPTION	Sets conditions for simulation.
	.IC or .NODESET	Initial values in circuit and subcircuit.
	.PARAM	Sets parameter values in the netlist.
	.GLOBAL	Sets the node name globally in the netlist.
Sources	Sources and digital inputs	Sets input stimuli (I or V element).
Netlist	.SUBCKT, .ENDS, or .MACRO, .EOM	Circuit elements for simulation. Subcircuit or macro definitions.
Analysis	.DC, .TRAN, .AC, and so on.	Statements to perform analyses.
	.SAVE and .LOAD	Saves and loads operating point information. (PrimeSim HSPICE only)

Table 5 Sections of the Input Netlist File (Continued)

Section	Command/Control Option Used	Description
Library, model and file Inclusion	.DATA	Creates a table for data-driven analysis.
	.TEMP	Sets temperature analysis.
	.INCLUDE	General include files.
	.MODEL	Element model descriptions.
	.LIB	Model library.
	.MALIAS	Assigns an alias to a diode, BJT, JFET, or MOSFET.
	.OPTION SEARCH	Search path for libraries and included files. (PrimeSim HSPICE only)
Output	.PRINT, .PROBE	Statements to output variables.
	.MEASURE	Statement to evaluate and report user-defined functions of a circuit.
Alter blocks	.ALTER	Sequence for inline case analysis.
	.DEL LIB	Removes previous library selection.
End of netlist	.END	(Required) End of netlist .

Examples

Table 6 Netlist Command Examples

Item	Example
Title	The first line is always the title of the netlist.
Comment characters	*: Comment character for a line
	\$. Comment character used after a command
Options	.option post
	.option runlvl=5
Print/Probe/Analysis	.print tran v(d) i(r1)

Table 6 *Netlist Command Examples (Continued)*

Item	Example
	<code>.probe tran v(g)</code>
	<code>.tran 0.1n 10n</code>
Initial conditions	<code>.ic v(b) = 0 \$ input state</code>
Sources	<code>Vg g 0 pulse 0 1 0 0.1n 0.1n 1n 2n</code> * example of a voltage source
Circuit description	<code>MN d g gnd n nmos</code> <code>RL vdd d 1K</code>
Models	<code>.model n nmos level = 49</code> <code>+ vto = 1 tox = 7n</code> * '+' continuation character
End	<code>.end \$ terminates the simulation</code>

Output Files

Table 7 lists the various types of output files produced by the PrimeSim HSPICE tool.

For information on the output files produced when performing advanced analog analyses, see PrimeSim HSPICE Advanced Analog Output Files in *PrimeSim HSPICE® User Guide: Advanced Analog Simulation and Analysis*.

Table 7 PrimeSim HSPICE Output Files and Their Extensions

Output File Type	Extension
AC analysis measurement results	.ma# ¹
AC analysis results (from .POST statement)	.ac#
Monte Carlo results	.mc#
Data-mining results	.mpp0
DC analysis measurement results	.ms#
DC analysis results (from .POST statement)	.sw#
FFT analysis graph data (from .FFT statement)	.ft#
Operating point information (from .OPTION OPFILE statement)	.dp#
Operating point node voltages (initial conditions)	.ic#
Output listing	.lis or user-specified
Output status	.st#
Output tables (from .DCMATCH OUTVAR statement)	.dm#
Pole-zero analysis results	.pz#
StatEye analysis measurement results	.mste#
StatEye analysis initial transient measurement results	.mtNp{f}# ²
StatEye analysis initial transient results	.trNp{f}# ³
StatEye analysis time-based results (from .POST statement)	.stet#
StatEye analysis voltage based results (from .POST statement)	.stev#

1. The hash character (#) can be either a sweep number or a hardcopy file number.
For .ac#, .dp#, .dm#, .ic#, .st#, .sw#, and .tr# files, # can be a number from 0 through 9999.
2. N is the incident port index, f is the edge index when edge is greater than 1, and # is the alter index.
3. N is the incident port index, f is the edge index when edge is greater than 1, and # is the alter index.

Table 7 *PrimeSim HSPICE Output Files and Their Extensions (Continued)*

Output File Type	Extension
Subcircuit cross-listing	.pa#
Transient analysis measurement results	.mt#
Transient analysis results (from .POST statement)	.tr#
Waveform files from .OPTION WDF argument for use with Synopsys WaveView/SX tools	*_wdf.tr#, *_wdf.sw#, or *_wdf.ac#

Simulation Examples

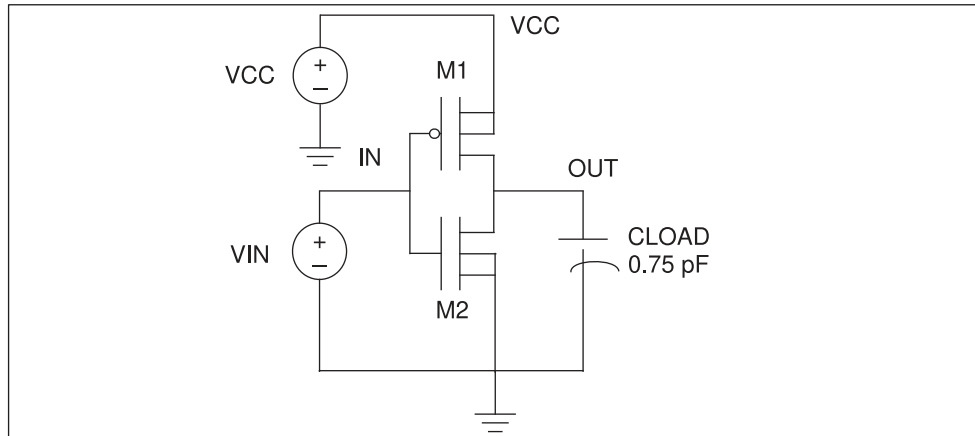
The following examples illustrate the basic PrimeSim HSPICE simulation types: DC, AC, and transient analysis:

- [DC Analysis of an Inverter](#)
- [AC Analysis of an RC Network](#)
- [Transient Analysis of an RC Network](#)
- [Transient Analysis of an Inverter](#)

DC Analysis of an Inverter

You can analyze the DC behavior of the simple MOS inverter shown in the [Figure 2](#) :

Figure 2 *MOS Inverter Circuit*



To analyze the DC behavior:

1. Type the following netlist data into a file named `quickDC.sp`:

```
Inverter Circuit
.OPTION POST
.DC VIN 0 5 0.1
.PRINT DC V(IN) V(OUT)
M1 OUT IN VCC VCC PCH L=1U W=20U
M2 OUT IN 0 0 NCH L=1U W=20U
VCC VCC 0 5
VIN IN 0 0 PULSE .2 4.8 2N 1N 1N 5N 20N
CLOAD OUT 0 .75P
.MODEL PCH PMOS LEVEL=1
.MODEL NCH NMOS LEVEL=1
.END
```

You can find the complete netlist for this example in `$installdir/demo/hspice/apps/quickDC.sp`.

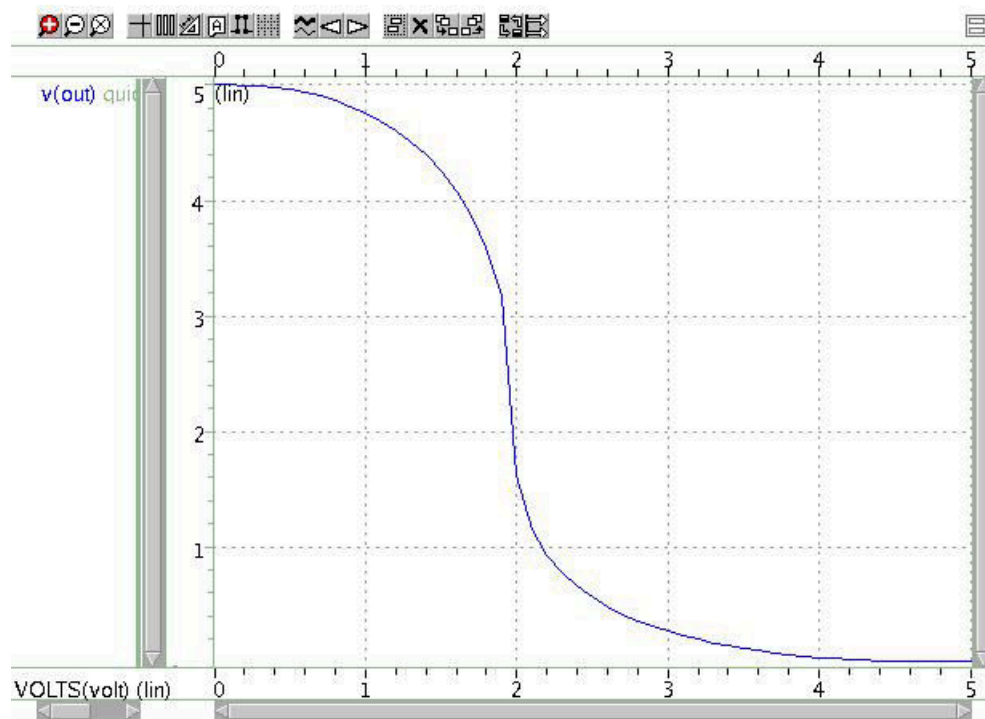
2. Run the PrimeSim HSPICE analysis by typing the following command:

```
% hspice -i quickDC.sp -o quickDC.lis
```

3. Use WaveView to examine the voltage waveform at the inverter OUT node.

[Figure 3 on page 38](#) shows the waveforms.

Figure 3 Voltage at Inverter Node v(out)

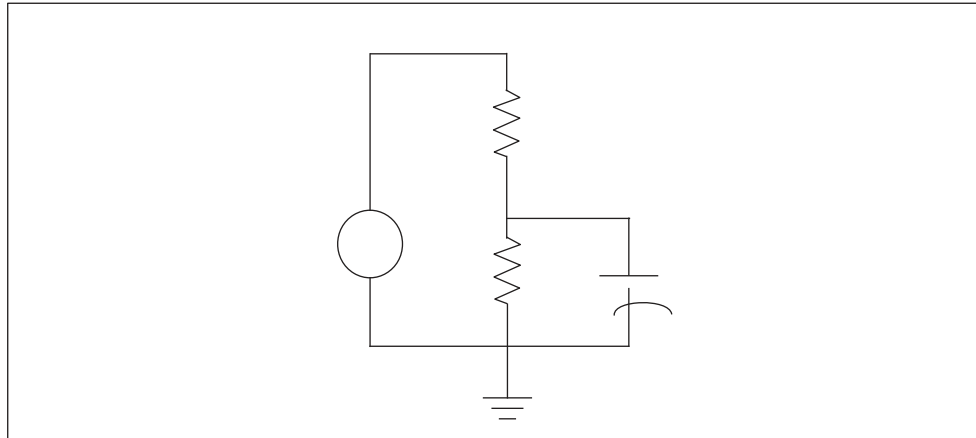


AC Analysis of an RC Network

[Figure 4 on page 39](#) shows a simple RC network with a DC and an AC source applied. The circuit consists of:

- Two resistors, R1 and R2
- Capacitor C1
- Voltage source V1
- Node 1 is the connection between the source positive terminal and R1
- Node 2 is where R1, R2, and C1 are connected
- PrimeSim HSPICE ground is always node 0

Figure 4



The netlist for this RC network is based on the demonstration netlist `quickAC.sp`, which is available in the directory `$installdir/demo/hspice/apps`:

```
A SIMPLE AC RUN
.OPTION LIST NODE POST
.OP
.AC DEC 10 1K 1MEG
.PRINT AC V(1) V(2) I(R2) I(C1)
V1 1 0 10 AC 1
R1 1 2 1K
R2 2 0 1K
C1 2 0 .001U
.END
```

To perform an AC analysis for an RC network circuit:

1. Type the above netlist into a file named `quickAC.sp`.
2. To run a PrimeSim HSPICE analysis, type:

```
% hspice quickAC.sp > quickAC.lis
```

When the run finishes, the PrimeSim HSPICE tool displays:

```
>info:      ***** hspice job concluded
```

This is followed by a line that shows the amount of real time, user time, and system time needed for the analysis.

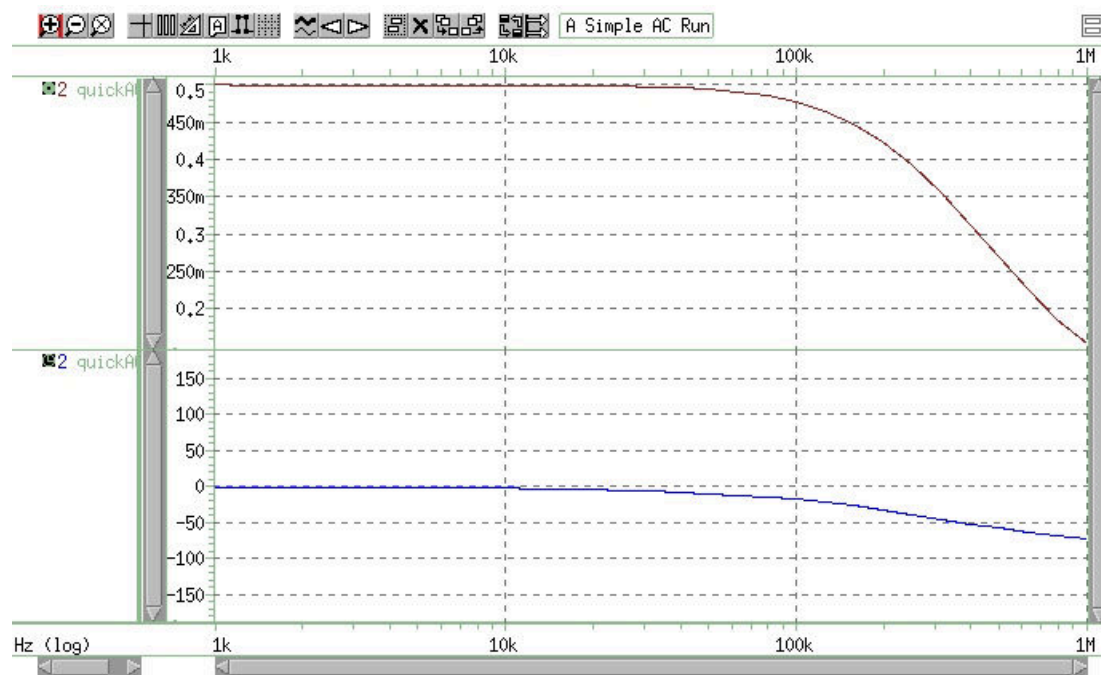
Your run directory includes the following new files:

- quickAC.ac0
- quickAC.ic0
- quickAC.lis
- quickAC.st0

3. Use a text editor to view the .lis and .st0 files to examine the simulation results and status.
4. Run WaveView.
5. From the menu bar, select **File > Import > Waveform File**.
6. Select the quickAC.ac0 file from the Open: Waveform Files window.
7. Display the voltage at node 2 by using a log scale on the x-axis.

Figure 5 shows the waveform that the PrimeSim HSPICE tool produces if you sweep the response of node 2, as you vary the frequency of the input from 1 kHz to 1 MHz.

Figure 5 RC Network Node 2 Frequency Response



As you sweep the input from 1 kHz to 1 MHz, the `quickAC.lis` file displays:

- Input netlist
- Details about the elements and topology
- Operating point information
- Table of requested data

The `quickAC.ic0` file contains information about the DC operating point conditions. The `quickAC.st0` file contains information about the simulation run status.

To use the operating point conditions for subsequent simulation runs, execute the `.LOAD` statement.

Transient Analysis of an RC Network

To run a transient analysis of an RC network with a pulse source, a DC source, and an AC source:

1. Type the following netlist into a file named `quickTRAN.sp`:

```
A SIMPLE TRANSIENT RUN
.OPTION LIST NODE POST
.OP
.TRAN 10N 2U
.PRINT TRAN V(1) V(2) I(R2) I(C1)
V1 1 0 10 AC 1 PULSE 0 5 10N 20N 20N 500N 2U
R1 1 2 1K
R2 2 0 1K
C1 2 0 .001U
.END
```

This example uses the demonstration netlist `quickTRAN.sp`, which is available in the directory `$installdir/demo/hspice/apps`.

Note:

The V1 source specification includes a pulse source. For the syntax of pulse sources and other types of sources, see *Sources and Stimuli* in the *PrimeSim HSPICE User Guide: Elements*.

2. To run PrimeSim HSPICE, type the following:

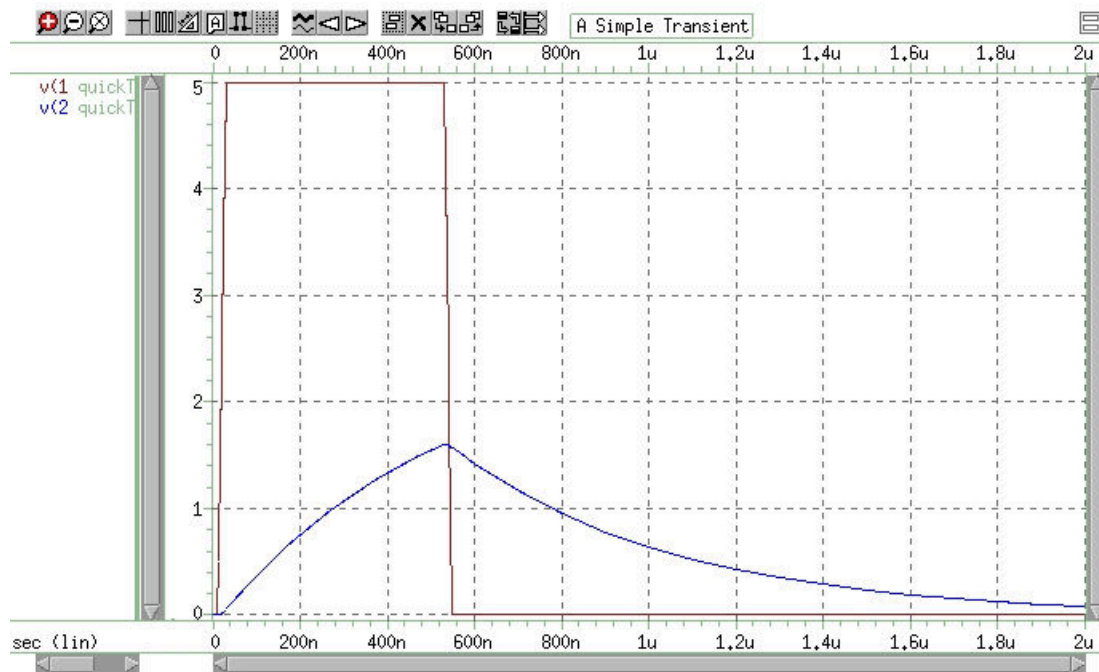
```
hspice quickTRAN.sp > quickTRAN.lis
```

3. To examine the simulation results and status, use a text editor and view the `.lis` and `.st0` files.
4. Run WaveView and open the `.sp` file.

5. From the menu bar, select **File > Import Waveform > File**.
6. Select the `quickTRAN.tr0` file from the Open: Waveform Files window.
7. Display the voltage at nodes 1 and 2 on the x-axis.

Figure 6 shows the waveforms.

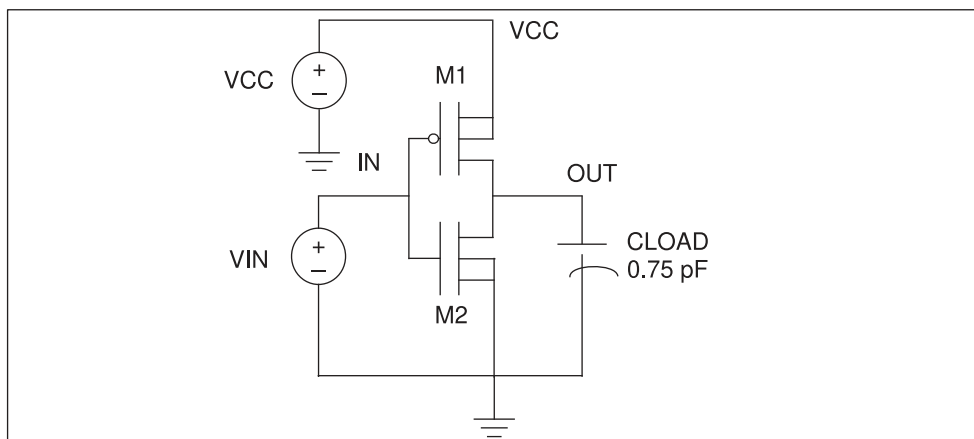
Figure 6 Voltages at RC Network Circuit Node 1 and Node 2



Transient Analysis of an Inverter

You can analyze the transient behavior of the simple MOS inverter shown in [Figure 7](#).

Figure 7 MOS Inverter Circuit



To analyze this behavior:

1. Type the following netlist data into a file named `quickINV.sp`:

```
Inverter Circuit
.OPTION LIST NODE POST
.TRAN 200P 20N
.PRINT TRAN V(IN) V(OUT)
M1 OUT IN VCC VCC PCH L=1U W=20U
M2 OUT IN 0 0 NCH L=1U W=20U
VCC VCC 0 5
VIN IN 0 0 PULSE .2 4.8 2N 1N 1N 5N 20N
CLOAD OUT 0 .75P
.MODEL PCH PMOS LEVEL=1
.MODEL NCH NMOS LEVEL=1
.END
```

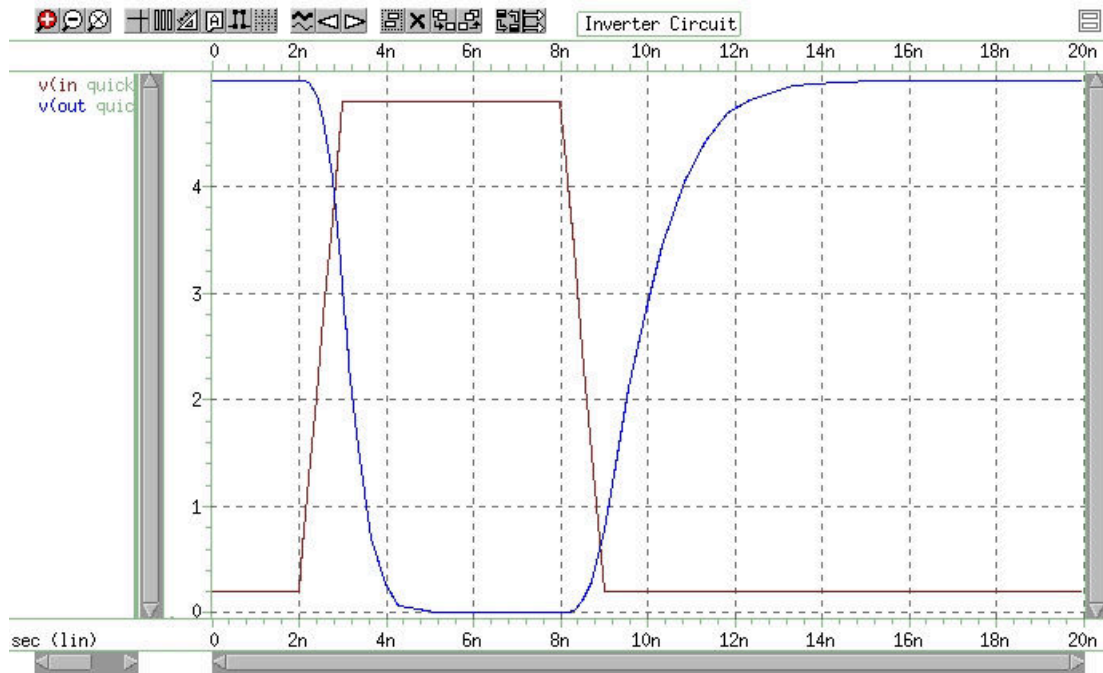
You can find the complete netlist for this example in the directory `$installdir/demo/hspice/apps/quickINV.sp`.

2. To run PrimeSim HSPICE, type:

```
hspice quickINV.sp > quickINV.lis
```

3. Use WaveView to examine the voltage waveforms, at the inverter IN and OUT nodes. [Figure 8 on page 44](#) shows the waveforms.

Figure 8 Voltage at MOS Inverter Nodes $v(in)$ and $v(out)$



4

Working With Elements

Presents an overview of elements in the PrimeSim HSPICE tool.

This chapter discusses the following topics:

- [Introduction to Elements](#)
- [About Element Instances](#)

Introduction to Elements

Element statements in the PrimeSim HSPICE tool describe the devices and sources in the netlist. Nodes are used to connect elements to one another. Nodes can be defined using either numbers or names.

Element statements specify the following characteristics of elements:

- The type of device
- The nodes to the connected device
- The operating electrical characteristics of the device

Element statements can also reference model statements that define the electrical parameters of the element.

For detailed descriptions of element statements for each of the supported elements, see *PrimeSim HSPICE® User Guide: Elements*. See also [Elements on page 14](#) in this guide.

About Element Instances

The names of element instances begin with the element key letter (see [Table 8](#)), except in subcircuits where instance names begin with X. Instance names can be up to 1024 characters long.

Table 8 *Element Identifiers*

Key Letter (First Character)	Element	Example Line
B	IBIS buffer	b_io_0 nd_pu0 nd_pd0 nd_out nd_in0 nd_en0 nd_outofin0 nd_pc0 nd_gc0
C	Capacitor	Cbypass 1 0 10pf
D	Diode	D7 3 9 D1
E	Voltage-controlled voltage source	Ea 1 2 3 4 K
F	Current-controlled current source	Fsub n1 n2 vin 2.0
G	Voltage-controlled current source	G12 4 0 3 0 10
H	Current-controlled voltage source	H3 4 5 Vout 2.0
I	Current source	I A 2 6 1e-6
J	JFET or MESFET	J1 7 2 3 GAASFET
K	Linear mutual inductor (general form)	K1 L1 L2 1
L	Linear inductor	LX a b 1e-9
M	MOS transistor	M834 1 2 3 4 N1
P	Port	P1 in gnd port=1 z0=50
Q	Bipolar junction transistor	Q5 3 6 7 8 pnp1
R	Resistor	R10 21 10 1000
S	S-parameter	S1 nd1 nd2 s_model2
V	Voltage source	V1 8 0 5
T, U, W	Transmission line	W1 in1 0 out1 0 N=1 L=1
X	Subcircuit call	X1 2 4 17 31 MULTI WN=100 LN=5

5

Working With Models

Presents an overview of the usage of standard device models in the PrimeSim HSPICE tool.

This chapter discusses the following topics:

- [Introduction to Models](#)
- [Selecting Models](#)

Introduction to Models

Every device model is a template defining various versions of each supported element type used in a netlist formatted for use by the PrimeSim HSPICE tool. Individual elements in your netlist can refer to these standard models for their basic definitions. When you use these models, you can quickly and efficiently create a netlist and simulate your circuit design.

Within your netlist, each element that refers to a model is known as an instance of that model. When your netlist refers to predefined device models, you reduce both the time required to create and simulate a netlist and the risk of errors, as compared to completely defining each element in your netlist.

See also [Models on page 15](#) in this guide.

Selecting Models

To specify a device in your netlist, use both an element and a model statement. The element statement uses the model name of the simulation device to reference the model statement. The following example uses the name PCH to refer to a MOSFET model. The example uses a PMOS model type to describe a P-channel MOSFET transistor:

```
M3 3 2 1 0 PCH L=1u W=1u
.MODEL PCH PMOS VERSION = 3.2 tnom=27.0 tox=1.00000E-08
+ xj=1.00000E-07 lint=8.195860E-08 wint=-1.821562E-07 vth0=-.86094574
+ vsat=60362.05
```

You can specify parameters in both element and model statements. If you specify the same parameter in both an element and a model, the element parameter (local to the specific instance of the model) always overrides the model parameter (global default for all instances of the model, if you do not define the parameter locally). The model statement specifies the type of device, for example, a MOSFET; the device type might be N-channel or P-channel.

Models can be selected from model libraries using the `.LIB` command. The following example calls the model library file `mosfet.lib`, which contains the PCH model used in the netlist. The model corner used in the netlist is the *tt* corner:

```
.LIB '../models/mosfet.lib' tt  
M3 3 2 1 0 PCH L=1u W=1u
```

A

Measurement System, Units, and Numbers

Describes the measurement system used in the PrimeSim HSPICE tool.

This appendix covers the following topics:

- [About the PrimeSim HSPICE Measurement System](#)

About the PrimeSim HSPICE Measurement System

The PrimeSim HSPICE tool uses the MKS (meter, kilogram, and second) measurement system, unless otherwise stated. The tool expects length and width in units of meters. However, the PrimeSim HSPICE tool will directly support the unit *mil* (=0.001 inch or 25.4e-06 meters) as input.

Caution:

Be careful when mixing units of measurements; otherwise, the PrimeSim HSPICE tool may not produce the results you expect.

You can enter numbers as an integer, a floating point, a floating point with an integer exponent, or an integer or a floating point with one of the scale factors listed in the [Table 9](#).

Table 9 Scale Factors

Scale Factor	Prefix	Symbol	Multiplying Factor
T	tera	T	1e+12
G	giga	G	1e+9
ME, MEG, X, or Z	mega	M, ME, X, or Z	1e+6
K	kilo	k	1e+3
MI or MIL	n/a	MI or MIL	25.4e-6
U	micro	μ	1e-6
N	nano	n	1e-9

Table 9 Scale Factors (Continued)

Scale Factor	Prefix	Symbol	Multiplying Factor
P	pico	p	1e-12
F	femto	f	1e-15
A	atto	a	1e-18
DB	DB	db	$10^{(\text{value}/20)}$
MIN	MIN	min	60
HR	HR	hr	3600
DAY	DAY	day	86400
YR	YR	yr	31536000

Note:

- Scale factor A is not a scale factor in a character string that contains *amps*. For example, the PrimeSim HSPICE tool interprets the string *20amps* as 20 amperes (of current), not as *20e-18 amps*.
- The scale factor M indicates either the suffix *milli* or *mega*. The PrimeSim HSPICE tool uses MEG or X to represent the suffix *mega* and M to represent the suffix *milli*. That is, 1m = 1e-3 (*milli*) and 1meg = 1x = 1e6 (*mega*).

B

Best Practices

Lists a few best practices for PrimeSim HSPICE netlists, netlist topologies, and analysis.

This appendix covers the following topics:

- [Netlists](#)
- [Netlist Topologies](#)
- [Analysis](#)

Netlists

Best practices for generating input netlists for the PrimeSim HSPICE tool include:

- Use either a schematic netlister or a text editor to generate the input netlist and library input files for the PrimeSim HSPICE tool.
- Each netlist line (logical record) cannot exceed 1024 characters. Use the + line continuation character to break up lines longer than 1024 characters, to avoid generating an error.
- An input file name can be up to 1024 characters long for all platforms, except Windows platforms which have a limitation of only up to 256 characters.
- The PrimeSim HSPICE tool has a limitation on the number of characters in a path name plus file name, of up to 1024 characters (except Windows platforms which have a limitation of only up to 256 characters). For example:

```
hspice -i path_name/input_file -o out_file
```

- When specifying a path and file name using `-i` or `-o`, the length must be 1024 characters or fewer, on all platforms. If the working directory path is greater than 1024 characters, the PrimeSim HSPICE tool aborts with an error message.
- Model names in a netlist must begin with a letter. If you enter a model name with a leading integer, the PrimeSim HSPICE tool issues a parsing error.

- Statements in the input netlist file can be in any order, except the first line which is the netlist title line. In the PrimeSim HSPICE tool, the last `.ALTER` submodule must appear at the end of the file and before the `.END` statement. If you do not place an `.END` statement at the end of the input netlist file, the PrimeSim HSPICE tool issues an error message.
- To indicate the ground node, use either the number 0, or the name GND, or GND!, or GROUND. The ground node is global.
- The PrimeSim HSPICE tool ignores differences between uppercase and lowercase characters in input lines, except in quoted filenames and after the `.INC` and `.LIB` commands. Use the `-case` command-line option if case sensitivity is required.
- Lines can be continued using the `+` character at the beginning of the continued line. For example:

```
V1 1 0 DC=0
+ PULSE (0 1 0 1n 1n 5n 10n
```

- Use a double backslash (`\\`) at the end of the line to continue the line on to the next line, when the continuation is inside a string. Whitespace is optional to precede the string continuation. For example:

```
g1noise 1 2 noise='sqrt(4*1.3806266e23*\\
(TEMPER+273.15)*0.001) '
```

Netlist Topologies

Best practices for constructing netlist topologies for the PrimeSim HSPICE tool include:

- When constructing a netlist, avoid the following topologies because they will cause the PrimeSim HSPICE tool to abort:
 - Voltage loops (that is, voltage sources in parallel with no other elements).
 - An ideal voltage source in a closed inductor loop.
 - Stacked current sources (that is, current sources in series).
 - An ideal current source in a closed capacitor loop.
- The PrimeSim HSPICE tool terminates floating power supply nodes with a 1 Meg Ω resistor and a warning message.
- Every node should have at least two connections, except for transmission line nodes (unterminated transmission lines are permitted) and MOSFET substrate nodes (which have two internal connections).

Analysis

Best practices for analysis statements for the PrimeSim HSPICE tool include:

- For the PrimeSim HSPICE tool to create waveform files, include `.option POST` or `.option WDB` in the input netlist file.
- If you are performing multiple analyses, to avoid warning messages from the PrimeSim HSPICE tool, set the analysis type in all `.PROBE` or `.PRINT` statements.
- Be careful while adding analysis statements in `.ALTER` blocks. The added analysis statement does not replace the analysis statement previously defined in the top level. Instead, the PrimeSim HSPICE tool executes the added analysis command in each `.ALTER` run, in addition to the analysis statement in the top level. This means that, the PrimeSim HSPICE tool will output more analysis result files than expected.
- When using temperature analysis with `.TEMP` and `.ALTER` blocks, it is recommended that the `.TEMP` value be parameterized and only one `.TEMP` statement at the top level be used. You can then change the temperature by changing the value of the parameter assigned to the `.TEMP` command.
- By default, the PrimeSim HSPICE tool uses the last parameter or option found in the netlist. Avoid multiple definitions of the same parameter or option.

C

Abbreviations and Acronyms

Lists the abbreviations and acronyms used in PrimeSim HSPICE documentation.

This appendix covers the following topics:

- [List of Abbreviations and Acronyms](#)

List of Abbreviations and Acronyms

Table 10 Abbreviations and Acronyms

Abbreviation/Acronym	Expansion
BER	Bit error rate
CFL	Compiled Function Library
CMI	Common Model Interface
DFT	Discrete Fourier transform
DP	Distributed processing
FFT	Fast Fourier transform
HSPUI	PrimeSim HSPICE graphical user interface
LIN	Linear network
LSF	Load Sharing Facility
LSTB	Loop stability
MOSRA	MOSFET reliability analysis
MT	Multithreading
SI	Signal integrity

Table 10 Abbreviations and Acronyms (Continued)

Abbreviation/Acro nym	Expansion
SOA	Safe Operating Area
SPUTIL	S-parameter Utility
StatEye	Statistical eye diagram